Statistics & Probability

Section 1: Permutations & Combinations

1 In how many ways can a president and a vice president be chosen from a committee of 6 people?

a. 30

b. 15

c. 36

d. 12

2 How many distinct committees of 3 students can be selected from a group of 7 students?

a. 35

b. 210

c. 21

d. 42

3 What is the value of $\frac{6!}{4! \times 2!}$?

a. 15

b. 30

c. 12

d. 20

Section 2: Conditional & Compound Probability

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4 If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, what is $P(A \cup B)$?

a. 0.8

b. 0.7

c. 0.9

d. 1.1

5 If $P(B) = 0.4$ and $P(A \cap B) = 0.12$, what is the conditional probability $P(A|B)$?

a. 0.30

b. 0.48

c. 0.25

d. 0.52

6 If events A and B are independent with $P(A) = 0.3$ and $P(B) = 0.4$, what is $P(A \cap B)$?

a. 0.12

b. 0.70

c. 0.10

d. 0.24

Section 3: Normal Distribution & Empirical Rule

7 A normal distribution has mean $\mu = 50$ and standard deviation $\sigma = 5$. What percentage of data falls between 45 and 55?

a. 68%

b. 95%

c. 99.7%

d. 50%

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8 In a normal distribution with mean 100 and standard deviation 10, what is the probability that a value is greater than 120?

a. 2.5%

b. 5%

c. 16%

d. 0.15%

9 What is the z-score for a value $x = 85$ from a distribution with mean $\mu = 70$ and standard deviation $\sigma = 10$?

a. 1.5

b. 1.2

c. 2.0

d. 0.85

Section 4: Measures of Dispersion & Variance

10 If the variance of a data set is 36, what is its standard deviation?

a. 6

b. 18

c. 72

d. 12

11 If 5 is added to every value in a dataset, how does the standard deviation change?

a. It remains unchanged

b. It increases by 5

c. It increases by 25

d. It multiplies by 5

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12 What is the range of the data set: $12, 5, 23, 18, 9, 31, 14$?

a. 26

b. 25

c. 24

d. 31

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